

Public Shelter Internet

Portable internet connectivity for underground bomb shelters in Israel



The Problem

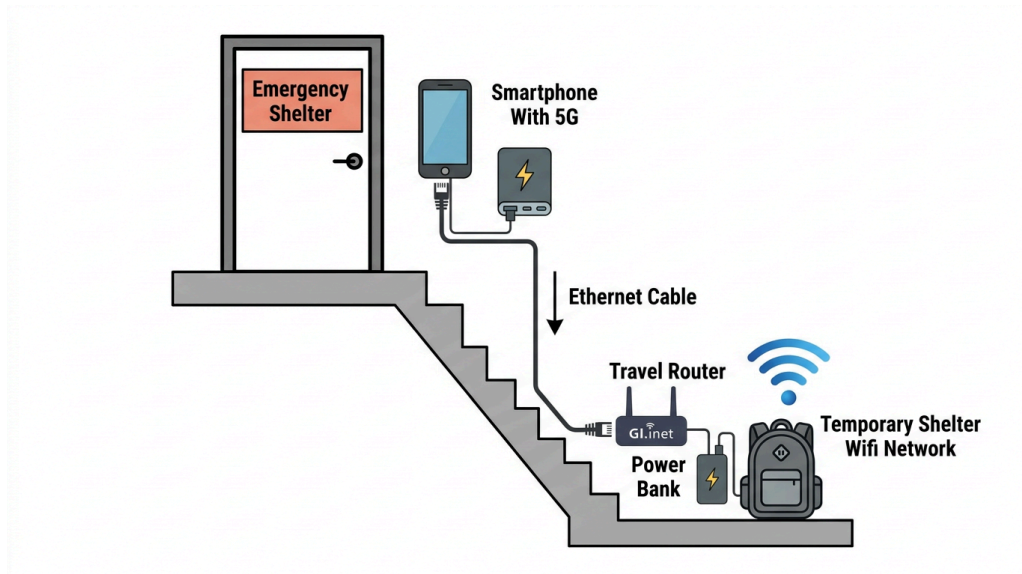
To be effective at stopping missiles, underground public shelters involve lots of concrete. Concrete does not love RF waves.

30% of Israelis do not have a protected space (*mamad*) in their homes. Israel relies primarily upon a smartphone app (Home Front Command) and wireless emergency alerts (WEA / cell broadcast) to send vital notifications to those sheltering in place.

Both of these notifications share a single point of failure: **internet connectivity**.

Many public shelters are dilapidated and besides lacking connectivity lack power outlets. People sheltering are often left to guess when an official all-clear has been issued by the Home Front Command. Unfortunately people tend to resort to guesstimates and leave shelter prematurely, risking shrapnel.

How It Works



Network topology: phone at shelter entrance, ethernet run to router inside

One Possible Solution

A backpack-portable kit that uses an Android phone's cellular connection, shared over a wired ethernet run into the shelter, and rebroadcast as Wi-Fi by a travel router.

Parts List

- An old but not too old Android phone (with 5G)
- A GL.iNet travel router (I use the GL-SFT1200 Opal)
- A USB-C to ethernet adapter (with USB-C to USB-A converter)
- A small power bank
- A 20M ethernet cable
- A zippered carry case
- Superglue
- USB switch (to turn the router on/off between uses)
- Velcro ties / cable ties

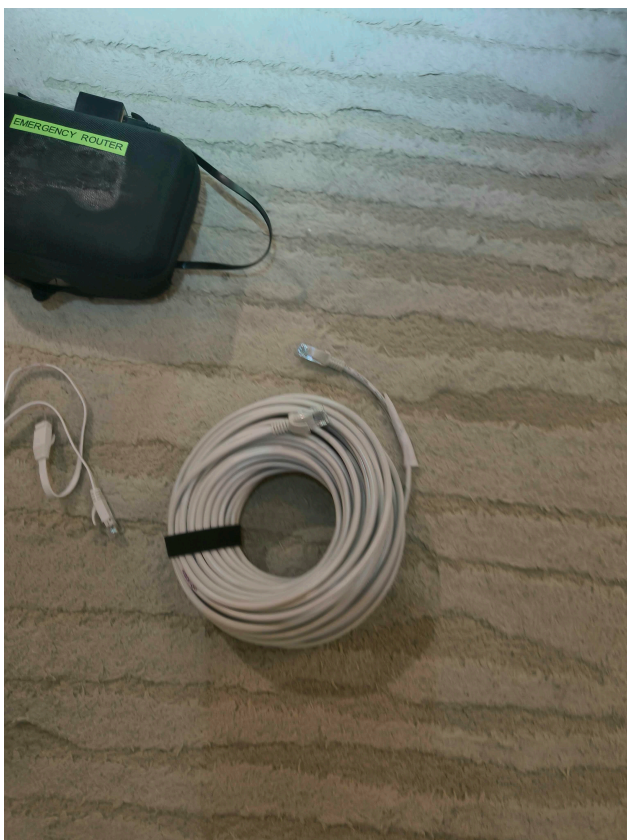
The Kit



Carry case, labeled and closed



Case open showing router and power bank



Full kit with 20M ethernet cable

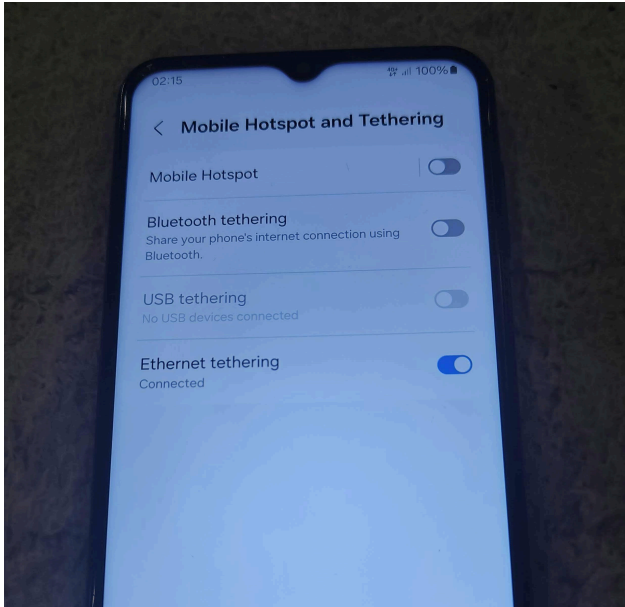


Everything fits in a backpack

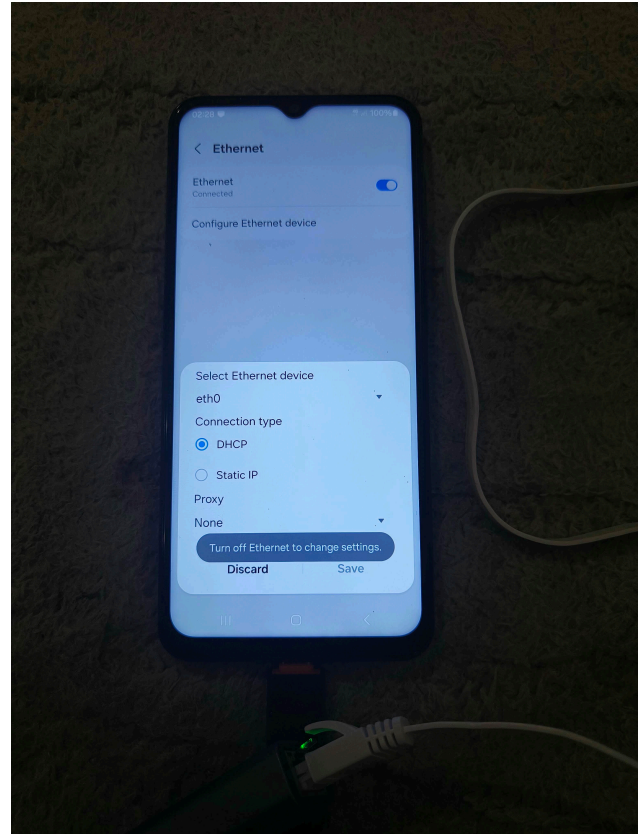
Setup

Phone Configuration

1. Connect the USB-C to ethernet adapter to the phone
2. Go to **Settings > Connections > Ethernet** and enable it
3. Set connection type to **DHCP**
4. Enable **Ethernet tethering** (Settings > Mobile Hotspot and Tethering)
5. **Disable Wi-Fi hotspot** — keep all bandwidth for the wired run



Enable ethernet tethering



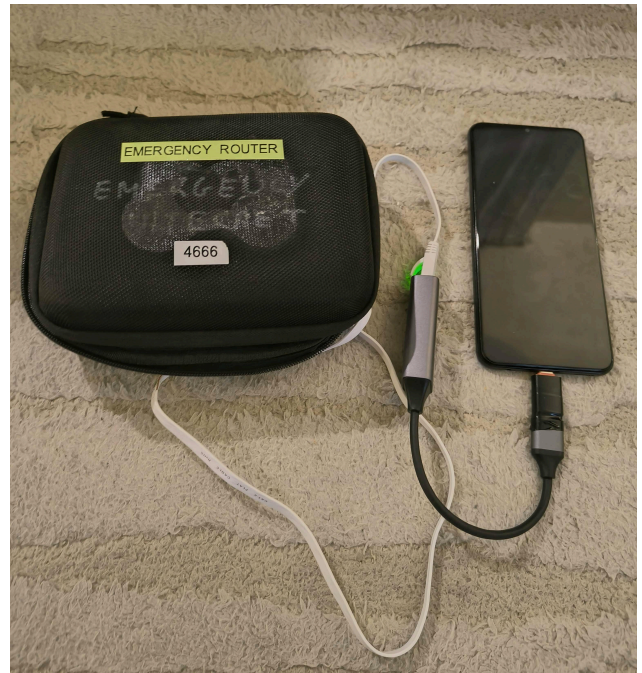
Ethernet with DHCP on eth0

Physical Setup

1. Position the Android phone near the top of the shelter stairs (best cellular signal)
2. Connect phone to ethernet adapter, run cable down the stairwell
3. Connect ethernet cable to the WAN port on the GL.iNet router
4. The router broadcasts a Wi-Fi network inside the shelter



Complete setup: phone to adapter to router



Case connected to phone via adapter

Simpler Setup (Shelter With Ethernet But No Wi-Fi)

Some newer or better-maintained shelters have an ethernet access point on the wall but still no Wi-Fi. In this case the setup is much simpler: just connect the GL.iNet router directly to the wall jack and it will broadcast Wi-Fi from the existing wired connection — no phone or tethering needed.



Router connected to shelter's ethernet jack



GL.iNet router powered by power bank

Laying the Ethernet Cable (No Existing Infrastructure)

You're not going to be running ethernet cable in an emergency. Nor do you want to create a trip hazard. Set this up **between alerts**.

- Secure the ethernet cable along the stairs/railings ensuring it doesn't impede the stairwell
- The carry case has an ethernet male-to-female coupler on the outside — connect the router when required
- If using an old Android you've accepted might be stolen: wipe it beforehand, set it up without apps/data, and leave it somewhere inconspicuous

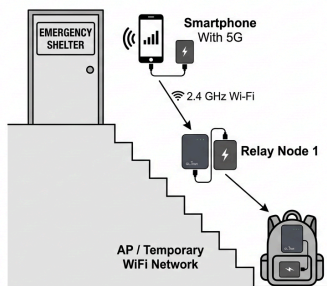
During an Alert

1. Grab backpack with internet kit
2. Plug the ethernet cable into the case coupler
3. Turn on the power bank (keep it charged between uses, turn it off after)
4. The router broadcasts Wi-Fi inside the shelter
5. When positioned right outside the shelter door, the signal usually propagates inside

Extending Coverage

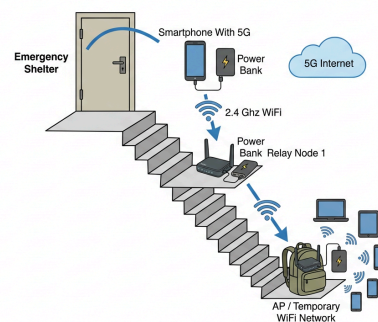
You can extend this significantly by adding additional GL.iNet travel routers in **repeater mode**:

Makeshift Mesh With Smartphone, One Relay, And AP
(2 x GL.iNet Routers + 2 X Power Banks)



Mesh setup with relay node

Makeshift Mesh With Smartphone, One Relay, And AP (2 x GL.iNet Routers + 2 x Power Banks)

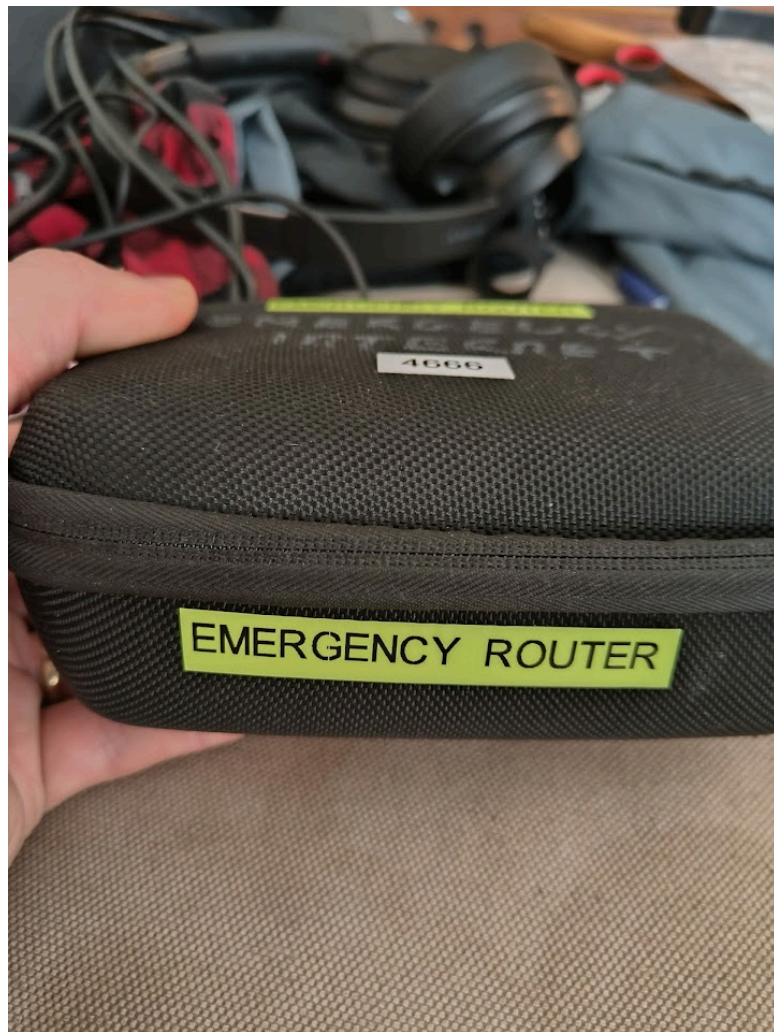


Detailed mesh topology

- Each repeater needs power (a power bank works; more advanced models need a PD-to-DC cable)
- Disable auto network switching on all nodes
- Hide the SSID on the first node (the phone) to prevent accidental connections
- Disable 5 GHz to save power
- Use a consistent, manually set Wi-Fi channel for reliable propagation between nodes

Build Tips

- Use superglue to attach the power bank and router together — you learn quickly when running between shelters that you want everything secure with no moving parts coming loose
- A USB switch lets you turn the router on/off without disconnecting anything
- Velcro ties keep cables managed inside the case



Emergency router case, ready to go